CREDIT RISK ANALYTICS DASHBOARD

Four-Framework Analysis Report

TOTAL APPLICANTS	TOTAL EXPOSURE	AT-RISK EXPOSURE	DEFAULT RATE
29,000	**$269M**	**$69M**	**22.50%**

Avg. Annual Income: $58.33K | Avg. Loan Amount: $9.25K | Avg. Interest Rate: 11.01%

Executive Summary

This report presents a comprehensive four-framework analysis of the Credit Risk Analytics Dashboard covering a portfolio of 29,000 loan applicants with $269M in total exposure. The analysis draws on key risk dimensions including debt-to-income ratio, loan-to-income ratio, annual income bands, loan grade classification, home ownership status, and loan purpose translating dashboard metrics into actionable business intelligence for senior stakeholders.

Key Portfolio Highlights

- 1 in 4 borrowers (22.5%) has defaulted representing $69M in at-risk capital.
- Loan Grade and repayment burden ratios are the dominant default drivers.
- Grade G borrowers default at 98.21% vs. only 10.51% for Grade A borrowers.
- Borrowers spending 30%+ of income on loan repayment default at nearly 6x the rate of low-burden borrowers.
- Lower-income borrowers (<$30K/yr) default at 4.4x the rate of high-income earners ($75K+).
- Renters represent the largest ownership segment (15.2K borrowers), requiring targeted risk monitoring.

Descriptive Analysis: What Is Happening?

Objective: Understand the current state of the portfolio through factual summary of key metrics.

Portfolio Overview

Total Applicants	29,000	Representative, large-scale portfolio
Total Loan Exposure	$269 Million	Significant capital at risk
At-Risk Exposure (Defaults)	$69 Million	25.7% of total exposure in default
Overall Default Rate	22.50%	1 in 4 borrowers has defaulted
Avg. Annual Income	$58,330	Mid-income borrower base
Avg. Loan Amount	$9,250	Moderate ticket size
Avg. Interest Rate	11.01%	Above-market prime rates

Borrower Distribution by Loan Grade

The portfolio is heavily concentrated in Grade A and B borrowers, which is expected for a health-scored lending institution. However, higher-grade segments carry significantly elevated risk:

Loan Grade	Borrower Count	Default Rate	Risk Level
A	9,650	10.51%	Low
B	9,306	16.95%	Moderate
C	5,825	21.67%	Elevated
D	3,219	60.21%	High
E	844	65.64%	Very High
F	204	72.06%	Critical
G	56	98.21%	Extreme

Borrower Distribution by Home Ownership

Renters dominate the portfolio at 15,200 borrowers (52.4%), followed by mortgage holders at 11,500 (39.7%), owner-occupiers at 2,300 (7.9%), and other categories at 100 (0.3%). This distribution signals that a majority of borrowers lack the asset stability that home ownership provides, which correlates to elevated default risk.

Loan Purpose Breakdown

Education leads all loan intents at 5,900 borrowers, followed by Medical (5,400), Venture (5,100), Personal (4,900), Debt Consolidation (4,700), and Home Improvement (3,100). The prevalence of education and medical loans suggests borrowers may face ongoing cash-flow constraints, contributing to repayment difficulty.

Conclusion

The portfolio presents a structurally sound volume with 29K applicants, but a 22.5% default rate signals systemic repayment stress. Grade A and B borrowers account for the bulk of volume but Grade D and above are disproportionate default contributors. The renter-heavy profile and mix of distress-driven loan purposes (medical, debt consolidation) reinforce vulnerability in the mid-to-lower income segment.

Diagnostic Analysis: Why Is It Happening?

Objective: Identify the root causes and key drivers behind the 22.5% default rate observed in the portfolio.

Debt-to-Income Ratio (DTI) as a Default Driver

DTI measures how much of a borrower's gross income is consumed by total debt obligations. The dashboard reveals a near-linear escalation in default risk as DTI increases:

DTI Band	Sample Count	Default Rate
0–20%	3,453	11.27%
20–30%	7,535	13.42%
30–40%	8,826	17.06%
40–50%	5,672	28.05%
50%+	3,618	56.66%

Once DTI crosses 50%, default risk quintuples vs. the safest cohort. This is the single strongest short-term predictor of distress high DTI borrowers are simply over-leveraged relative to income.

Loan-to-Income Ratio (LTI) as a Default Driver

LTI measures the size of the specific loan relative to annual income. The dashboard shows an even steeper relationship:

LTI Band	Sample Count	Default Rate	Risk Multiple
0–10%	7,922	12.03%	1.0x (Base)
10–20%	10,962	14.57%	1.2x
20–30%	6,318	21.60%	1.8x
30%+	3,902	67.45%	5.6x

Loans exceeding 30% of annual income carry a 67.45% default rate nearly 6x that of low-burden borrowers. This points to loan sizing as a critical underwriting failure point: borrowers may qualify on income but are unable to sustain repayment when the loan is too large relative to earnings.

Income Band as a Default Amplifier

Annual income shapes the borrower's fundamental ability to absorb financial shocks. Sub-$30K earners exhibit a 47.27% default rate 4.4x higher than $75K+ earners at 9.85%. The 30–50K income band is critical, housing the largest number of borrowers (18,274 total) at still-elevated default rates.

Income Band	Borrower Count	Default Rate	Risk Classification
< $30K	3,493	47.27%	Very High Risk
$30K – $50K	9,116	28.25%	Elevated Risk
$50K – $75K	9,158	17.45%	Moderate Risk
$75K+	7,337	9.85%	Low Risk

Loan Grade as a Systemic Risk Signal

Loan grades encapsulate the institution's internal credit scoring. The data validates the grade ladder as a reliable risk indicator: a near-monotonic escalation from A (10.51%) to G (98.21%). Grade D and above collectively represent the highest-risk cohort, and any relaxation of underwriting standards that allows volume growth in D-G territory will materially increase portfolio default rates.

Conclusion

The root causes of the 22.5% default rate are multi-layered but clearly linked to four compounding factors: (1) excessive debt burden relative to income (high DTI), (2) oversized loan amounts relative to income (high LTI), (3) low income base limiting repayment resilience, and (4) weaker credit profiles concentrated in Grade C-G segments. These factors compounda low-income borrower with a high LTI and a Grade D rating faces near-certain default based on portfolio evidence.

Predictive Analysis: What Will Happen?

Objective: Project future default outcomes and identify borrower segments most likely to default if portfolio composition remains unchanged.

Projected Default Risk by Portfolio Segment

Based on the empirical default rate relationships identified in the diagnostic phase, the following segments represent highest forward-looking default risk:

Segment / Risk Factor	Current Default Rate	Forward Risk Outlook	Priority Action
Grade D–G Borrowers	60–98%	Criticalnear-certain loss	Restrict new origination
DTI 50%+ Borrowers	56.66%	Will default at high rate if rate rises	Flag for early intervention
LTI 30%+ Borrowers	67.45%	Loan misalignmenthigh rollover risk	Reduce max loan ceiling
Income < $30K Borrowers	47.27%	Income shock sensitivity is extreme	Require income verification
Renters (No Asset Collateral)	Higher exposure	Rising rates increase strain	Consider collateral requirement

Portfolio Stress Scenarios

If macroeconomic conditions deteriorate (interest rate hikes, income shocks, or employment disruption), the following forward scenarios are projected:

- Scenario A Baseline (No Change): Default rate holds at 22.5%, with $69M at-risk exposure.
- Scenario B Rate Increase (+200bps): DTI bands shift up by 3–5%. Borrowers near the 40–50% DTI threshold migrate into the 50%+ band. Estimated default rate rises to 27–29%, increasing at-risk exposure to approximately $82–85M.
- Scenario C Income Shock (10% Wage Decline): Sub-$30K segment expands. Default rate in <$30K band could rise from 47% to 55%+. Portfolio-level default rate rises to approximately 25–26%.
- Scenario D Grade Drift (5% more D-G originations): Even a 5% increase in Grade D-G originations would push the portfolio default rate from 22.5% to approximately 25–27% due to the exponential risk premium of those grades.

Highest-Risk Borrower Profile

High-Risk Borrower Profile

Annual Income: < $30,000
Loan Grade: D, E, F, or G
DTI: 40% or above
LTI: 30% or above
Loan Purpose: Medical or Debt Consolidation
Home Ownership: Renter

A borrower matching all 6 criteria faces an estimated default probability exceeding 85% based on the compounding risk factors observed in the portfolio.

Predictive Analysis Conclusion

The portfolio is acutely vulnerable to any macroeconomic disruption. The Grade D-G cohort, DTI 50%+ borrowers, and sub-$30K income earners represent the primary default acceleration vectors. Without targeted intervention, an external stress event could push at-risk exposure from $69M to $85M+. Early warning systems and proactive outreach to high-risk segments are critical within the next 6–12 months.

Prescriptive Analysis: What Should We Do?

Objective: Translate insights into concrete, prioritized business actions to reduce default rates, protect capital, and optimize portfolio performance.

Immediate Actions (0–3 Months)

Action 1: Implement Hard Caps on LTI and DTI at Origination

- Cap new loan originations at LTI ≤ 25% (reduce from current uncapped state) to avoid the 30%+ band where defaults jump to 67.45%.
- Decline or escalate for manual review any applicant with DTI > 45% at application stage.
- Expected impact: Prevents the highest-risk lending bucket from growing; preserves capital.

Action 2: Suspend Grade F and G Originations

- Grade F (72.06% default) and Grade G (98.21% default) generate near-certain capital loss.
- Suspend new originations in these segments immediately pending a full underwriting review.
- Expected impact: Eliminates the highest default probability cohort from future exposure growth.
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Action 3: Flag & Triage Existing High-Risk Accounts

- Extract all current borrowers with DTI > 50% AND Grade D or above into a watch list.
- Initiate proactive outreach restructuring offers, payment deferrals, or hardship programsbefore accounts go delinquent.
- Expected impact: Early intervention reduces forced write-offs; recovery rates are significantly higher pre-default.

Medium-Term Actions (3–12 Months)

Action 1: Strengthen Income Verification for Sub-$30K Applicants

- Sub-$30K borrowers default at 47.27% nearly half the cohort. Introduce mandatory income verification (bank statements, pay stubs, tax returns) for all applicants below $35K annual income.
- Consider requiring a co-signer or collateral for loans > $5K in this income segment.
- Expected impact: Reduces adverse selection in the highest-risk income tier.

Action 5: Introduce Dynamic Interest Rate Pricing by Risk Grade

- Grade B–D borrowers currently share a blended rate near 11.01%. Introducing risk-adjusted pricing ensures high-risk borrowers pay rates that compensate for their default probability.
- Suggested bands: A = 7–9%, B = 9–11%, C = 11–14%, D = 14–18%, E+ = Decline or secured-only.
- Expected impact: Improves risk-adjusted return on the portfolio and acts as a natural deterrent to high-risk volume.

Action 6: Target Medical and Debt Consolidation Loan Applicants with Affordability Assessment

- These two purposes account for 10,300+ borrowers and are likely distress-driven, not growth-driven.
- Introduce a structured affordability assessment (post-loan cash flow analysis) to ensure borrowers have sufficient residual income after repayment.

Long-Term Strategic Actions (12–24 Months)

Action 1: Build a Real-Time Default Early Warning System

- Develop a model-driven early warning system that scores borrowers monthly on DTI, payment behavior, and income signals.
- Trigger automated outreach at a risk score threshold, reducing the time between risk signal and intervention.

Action 2: Shift Portfolio Mix Toward Grade A–B and $50K+ Income Segments

- Strategic origination targeting should increase the share of Grade A–B and $50K+ income borrowers to improve overall portfolio quality.
- Set a portfolio health KPI: Grade A–B share > 70% of new monthly originations.

Action 3: Consider Secured Lending Options for High-Risk Applicants

- For creditworthy but high-DTI borrowers, offer secured loan products (against vehicles, property, savings) that reduce loss-given-default even if probability of default remains elevated.

Conclusion

A three-horizon intervention plan immediate caps and suspensions, medium-term underwriting reforms, and long-term portfolio rebalancing can realistically reduce the default rate from 22.5% toward a target of 15–17% within 18 months. The estimated capital protection from immediate actions alone (LTI caps, Grade F/G suspension, and high-risk triage) could reduce at-risk exposure by $15–20M annually.

Overall Conclusion

The Credit Risk Analytics Dashboard reveals a portfolio under moderate-to-high systemic stress. While the institution maintains healthy volume with 29,000 applicants and $269M in exposure, the 22.5% default rate and $69M in at-risk exposure demand immediate strategic attention.

Analysis Type	Core Finding	Priority
Descriptive	22.5% default rate; Grade G at 98.21%; renters dominate at 52%	Understand
Diagnostic	High DTI (50%+), high LTI (30%+), and low income (<$30K) compound default risk	Explain
Predictive	Grade D-G + high DTI borrowers face near-certain default; macro shocks could lift exposure to $85M+	Anticipate
Prescriptive	Cap LTI/DTI, suspend Grade F-G, triage watch list, reform pricing, and shift portfolio mix	Act

The data is clear: the most impactful immediate lever is LTI and loan grade management at origination. The most powerful long-term lever is systematic portfolio rebalancing toward higher-income, lower-risk borrowers. Together, these interventions can meaningfully protect capital, reduce defaults, and position the institution for sustainable credit growth.